

HAT-P-55b

R-0009

Benchmark run · workflow W8-benchmark · record 2026-bench_HATP-55_260705 · created 2026-07-08T03:34:03+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461226.733 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.046
Transit depth (Rp/Rs)^2	2.59 +/- 1.5 [%]
Orbital Inclination (inc)	89.27 +/- 2.34
Airmass coefficient 1 (a1)	1.42 +/- 0.76
Airmass coefficient 2 (a2)	-0.49 +/- 1.26
Scatter in the residuals of the lightcurve fit is	53.24 %
Best Comparison Star	#2 - [604, 154]
Optimal Aperture	1.5
Optimal Annulus	14.0
Transit Duration (day)	0.135 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.161 ± 0.046 vs 0.1202 (deviation 0.89σ)
Duration fit vs published	0.135 d vs 0.1225 d (deviation 0.89σ)
Mid-time O–C	24.1 min
Depth SNR	3.5
Residual scatter	53.24 %

Light curve

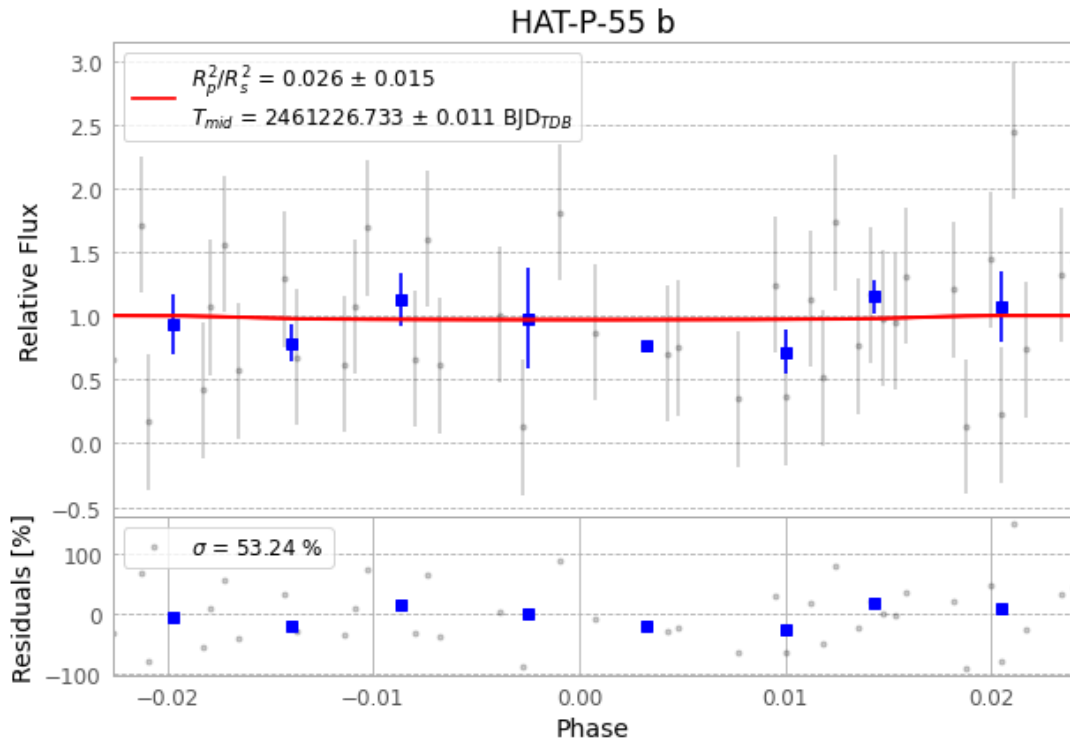


Figure 1 — FinalLightCurve_HAT-P-55 b_2026-07-04.png (SHA-256 5e44a232b211392b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [581, 415], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9f76818418e2d7a2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

93 FITS frames (61 MB), HATP-55260705025519.FITS → HATP-55260705073316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-55-260705.log (390ce58d14ee...), FinalLightCurve_HAT-P-55 b_2026-07-04.png (5e44a232b211...), FinalLightCurve_HAT-P-55 b_2026-07-04.csv (5e75f84cc774...)

Compute resources

Wall time 171.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.